



Internship Project Presentation on Predicting rate of Employee Attrition

Presented By:

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Job Profile

➤ Job Detail:

- Job Title: Python and Machine Learning
- Job position: Intern
- Company: Department of AI & DS, AI Club and Centre of Excellence, IGDTUW

➤ Job Responsibilities:

- The interns were required to attend online sessions and create projects using whatever we had learnt.
- We also had to attend sessions organised with industry professionals who told us about their work domain and experience.
- Minor and Major Projects were to be submitted.

Work Description

- Several technical sessions were conducted by experts from different academic institutions and industry with vast experience in AI and Machine Learning.
- The topic chosen by me for the minor project was “Honey Production Analysis”. I downloaded the freely available dataset from kaggle and performed data analysis on it and answered the questions based on it.
- The major project was based on the knowledge gained after studying various machine learning models. For major project, we made a group of three members and chose the topic “Employee Attrition”.
- We did data cleaning and data preprocessing, finally preparing our dataset for training and testing.
- We split our dataset into 70% training and 30% testing ratio and implemented 3 different ML models namely, Logistic Regression, Support Vector Machine and Random Forest Classifier.

Work Outcome

➤ Minor Project Outcome:

- Got to learn about python libraries like numpy, pandas, matplotlib.
- I stated conclusions and suggestions for increasing the honey production demand based on my analysis of the dataset.
- I was able to conclude that the total honey production has declined over the years and value of production per pound has increased.
- Reason :
There was a continuous decrease in the yield per colony over the years.

Link: <https://colab.research.google.com/drive/19jfV0XDcdHNIxUKT9LfOZYFUITAKplHr?usp=sharing>

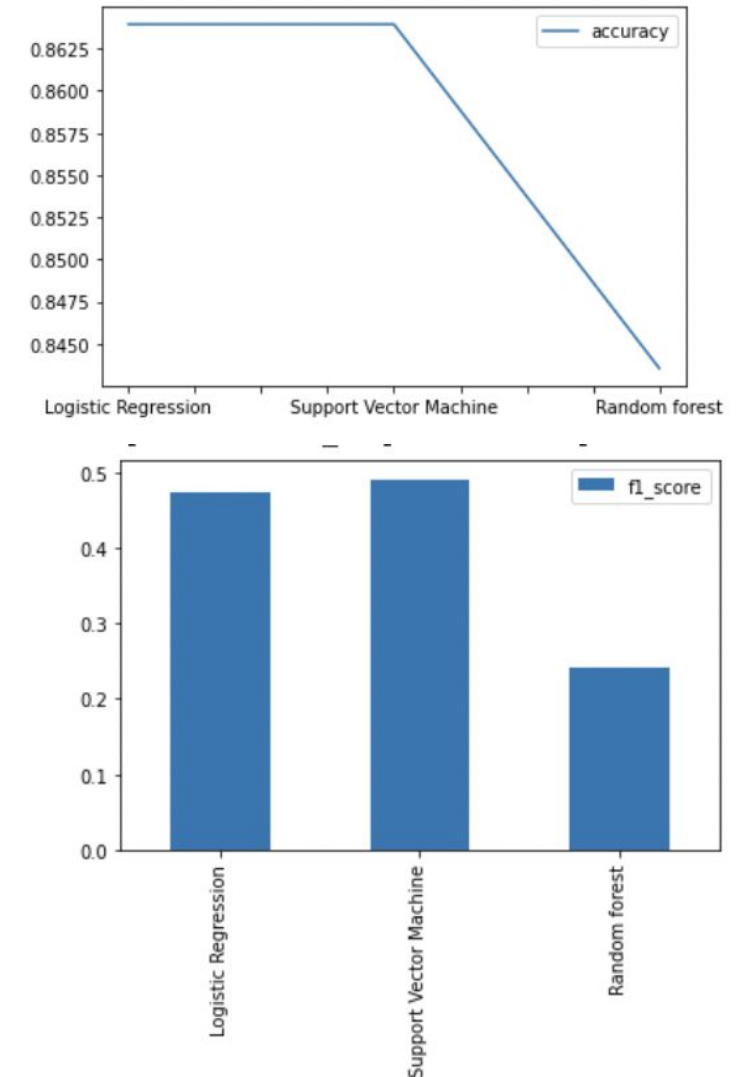
Work Outcome

➤ Major Project Outcome:

- Learnt about preprocessing of data
- Implemented three ML models on the given dataset which had 35 variables with 1470 data entries.
- Calculated the accuracy, f1 score of each model.
- The models Random Forest and Logistic Regression gave f1 scores of 0.241 and 0.473. These models gave an accuracy of 86.39% and 84.35%.
- From the analysis of the dataset and calculating accuracies and f1 score, we found SVM has the highest accuracy, meaning it guesses correctly **86.39%** of the predictions.
- We wrote a research paper at the end which is still under review by our mentor.
- Certificate was awarded after successful completion of the Internship.

Link:

<https://colab.research.google.com/drive/15jM5z2DHNKMkTbrD-qXYkXb9h0YOrIJJ?usp=sharing>



Research Paper

- In the research paper, an attempt has been made to analyse the publicly available employee attrition dataset using suitable models, study them and propose the best model on the basis of their prediction accuracies of employee attrition.
- The research paper is based on the major project done by me and my group members.
- Authors of the paper: Ananya Gupta, Pari Khaitan and Sakshi Sihag
- Link to the paper:
https://drive.google.com/file/d/1yeDC26mz7z4Vf7fM_f60FmJvG-WtICB2/view

Predicting rate of employee attrition using machine learning algorithms

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Abstract—Employees are the most valuable assets of the organization. Their presence adds value to the organization in terms of both quality and quantity. When employees leave the organization, they carry with them invaluable tacit knowledge which is often the source of competitive advantage for the business. Thus, business leaders sometimes have to face huge setbacks due to the loss of talented employees. Irrespective of the kind of industry or the structure of the business, attrition is a common problem in every organization which not only hampers production but also results in heavy long-run costs and loss of goodwill to the organization. In this paper, an attempt has been made to analyse the publicly available employee attrition dataset using suitable models, study them and propose the best model on the basis of their prediction accuracies of employee attrition.

Index Terms—attrition, machine learning, prediction

I. INTRODUCTION

Attrition is the departure of employees from the organization for any reason (voluntary or involuntary), including resignation, termination, death or retirement. It is an inevitable part of any business. Some forms of attrition are unavoidable, like if an employee is retiring or is moving to another city. But after a certain threshold, attrition can make a big dent in a company's bottom line as well as its culture. Over the years, maintaining a permanent and promising workforce has become a tough task for employers and thereby increased attrition in organizations. Therefore, attrition is an alarming concern for all organizations across various sectors. Anything that involves manpower would be affected by employee attrition. Therefore nowadays, the importance and relevance of the employees are well understood by organizations and hence the focus is on understanding the key factors that affect employee attrition. The prediction of employee attrition at multinational companies can be very crucial for its overall growth.

Therefore, we have applied various machine learning and deep learning algorithms to predict the rate of employee attrition.

The aim of this research paper is to analyse the employee attrition dataset using various machine learning models and algorithms like Logistic Regression, Support Vector Machine (SVM) and Random Forest Model. We have compared the accuracies and F1 scores using graphs, and have chosen the best model out of these, for the prediction of employee attrition.

II. LITERATURE SURVEY

Middle level officers are more likely to leave, may be due to some disagreement with their senior officer as proposed by [3]. They observed major factors that influenced employee abandonment from the firm. The two rules are moderately derived by him. Some set of questions are asked with the both parties and depending upon their answers he concluded some facts based on workload, objectives, carrier opportunity and firm management. Human resource management [4] endeavors on basically termination rates and dismissal rates but actual content of them are enormously different. The previous model shows that, there are several distinct levels of attrition and turnover. Some research dictates that the consequences of dismissal and termination rates are at organizational level. Allen and Meyer (1990) [5] described the three-basis entity for the negative side of the turnover.

Regulating officer will more probable leave from the organization because of a contention with the higher administration than a representative who is in struggle with his prompt director. He recognized the determinant figures that influence employee acceptance [5] without protest from the organization. Two arrangements of information social occasion techniques were directed. An equivalent number of representative and officer respondents were solicited to answer a set from polls that were ordered by workload, objectives, identity, professional success, and hierarchical administration. The after-effects of the two information gathering methods demonstrated that the most noteworthy component that adds

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